

Differential $\text{CC}\pi^\pm$ cross sections on argon in the NuMI beam

$\nu_\mu + \bar{\nu}_\mu$, FHC, RHC and combined; inclusive and proton-tagged

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Review panel, 30 September 2026

Analysis documents v1.6 (tag v1.6, 34baa99)

The signal region is blind. Every cross section shown is extracted from central-value pseudo-data.

The documents and this talk

Three documents (v1.6)

- ▶ **Analysis note** (54 pp.): measurement definition, selection, extraction, uncertainties, validation, inclusive results.
- ▶ **Proton-tagged note** (18 pp.): subsample with a reconstructed proton above $0.3 \text{ GeV}/c$; refers to the analysis note for everything shared.
- ▶ **Technical supplement** (54 pp.): final configuration, evidence for each choice, rejected alternatives.
- ▶ Data release: A_C , covariances by source and curves of 59 extractions.

Outline (order of the analysis note)

1. Measurement definition
2. Beam, flux, exposure, samples
3. Signal definition and selection
4. Backgrounds and control regions
5. Extraction: response, binning, Wiener-SVD
6. Systematic uncertainties
7. Validation
8. Inclusive results
9. Proton-tagged subsample
10. Limitations and summary

Backup: technical supplement.

Measurement definition

Interaction	ν_μ or $\bar{\nu}_\mu$ CC on argon; $\mu^- \pi^+$ and $\mu^+ \pi^-$ both signal (charge-blind)
Final state	Exactly one μ and one $\pi^\pm$ among primaries; no π^0 , kaon or other meson; any number of nucleons; coherent included
Phase space	$p_\mu > 0.15 \text{ GeV}/c$, $p_\pi > 0.175 \text{ GeV}/c$, $\theta_{\mu\pi} < 2.6 \text{ rad}$
Vertex	True vertex in FV $10 < x < 246$, $-101 < y < 101$, $10 < z < 986 \text{ cm}$, excluding $675.1 < z < 775.1 \text{ cm}$
Neutrino direction	Angles about the ν : true direction per event (truth), NuMI target→vertex (reco), 28.6° from detector z
Flux	$\Phi_{\nu_\mu} + \Phi_{\bar{\nu}_\mu}$, new-G4 PPFX CV, $E_\nu > 60 \text{ MeV}$: FHC 6.81159, RHC 6.44646, comb. $6.59691 \times 10^{-10} \text{ cm}^{-2} \text{ POT}^{-1}$
Exposure	FHC 7.766, RHC 11.082, combined $18.848 \times 10^{20} \text{ POT}$; each run normalised to its own exposure
Target	$N_{\text{Ar}} = 8.710 \times 10^{29}$ argon nuclei in the FV
Normalisation	$10^{-38} \text{ cm}^2/\text{Ar}$; $d\sigma/dx _j = \hat{N}_j/(\Delta x_j \Phi N_{\text{Ar}})$
Unfolding	Wiener-SVD; result estimates $A_C x_{\text{true}}$; predictions compared after A_C ; one-bin total has $A_C = 1$
Observables	p_μ (6), p_π (3 regions), $\cos \theta_\mu$ (8), $\cos \theta_\pi$ (5), $\theta_{\mu\pi}$ (5), one-bin total; FHC, RHC, comb. $\cos \theta_\pi \times \cos \theta_\mu$ (2×3), combined only
Covariance	Flux, interaction model, re-interaction, detector, μ momentum scale, POT, targets, MC/EXT/data statistics

Primary results: 15 differential extractions (5 observables $\times$ 3 configurations) and 3 one-bin totals.

Secondary result: $d^2\sigma/d\cos\theta_\pi d\cos\theta_\mu$, combined.
Proton-tagged results in the companion note.

Physics context

- ▶ $\text{CC}\pi^\pm$ is among the most frequent interactions at 1–5 GeV and a background to ν_e appearance (NOvA, T2K, DUNE).
- ▶ Dominated by $\Delta(1232)$; non-resonant + DIS $\approx 20\%$; coherent $\leq 10\%$ (signal).
- ▶ FSI (absorption 25–30%, charge exchange) moves events between $\text{CC}\pi^\pm$, $\text{CC}\pi^0$, $\text{CC}0\pi$ at the 30% level: the signal is defined by the final state.
- ▶ Response and pseudo-data from the MicroBooNE tune of GENIE v3.
- ▶ Standalone generators span a factor 1.6 in the total, GENIE lowest, NEUT highest.

Generator	Resonances	FSI
GENIE v3	KLN–BS, 18 res.	INTRANUKE hA
NuWro	Δ only	Salcedo–Oset cascade
NEUT	mod. R–S, 18 res.	step-by-step cascade
GiBUU	in-medium spectral fn.	transport, π optical pot.

Closest measurements

- ▶ MicroBooNE BNB $\text{CC}1\pi^\pm$:
 $\sigma = (3.75 \pm 0.80) \times 10^{-38} \text{ cm}^2/\text{Ar}$, $\langle E \rangle \approx 0.7 \text{ GeV}$.
- ▶ MicroBooNE NuMI $\nu_e \text{CC}\pi^+$: same beam, framework and conventions.
- ▶ The value here, $\approx 1.0 \times 10^{-38}$, is not directly comparable to BNB: the flux average runs from 60 MeV while pion production starts near 0.3 GeV.

NuMI flux and exposure

Integrated flux ($10^{-10} \text{ cm}^{-2} \text{ POT}^{-1}$, $E_\nu > 60 \text{ MeV}$)

	Φ_{ν_μ}	$\Phi_{\bar{\nu}_\mu}$	$\Phi_{\nu_\mu + \bar{\nu}_\mu}$	$\bar{\nu}_\mu$
FHC	4.43515	2.37644	6.81159	34.9%
RHC	2.92348	3.52298	6.44646	54.6%
Comb.	3.54634	3.05057	6.59691	46.2%

- ▶ Each configuration normalised with its own flux (RHC is 5.4% below FHC).
- ▶ Mean energy 0.48 GeV; only 40% of the flux lies above 0.3 GeV.
- ▶ PPFX CV weight on every event (mean 0.944 on selected signal); flux integrals validated four ways.
- ▶ Selected RHC events are 76% ν_μ : the ν_μ cross section is $2\text{--}3\times$ larger.

Exposure by run (10^{20} POT)

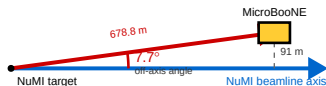
Run	MC POT	Data POT	Scale
Run 1 FHC	23.282	2.192	0.0942
Run 2 FHC	24.934	1.268	0.0509
Run 4 FHC	28.335	2.075	0.0732
Run 5 FHC	19.300	2.231	0.1156
FHC	95.85	7.766	per run
Run 1 RHC	8.997	0.605	0.0673
Run 2 RHC	57.865	2.591	0.0448
Run 3 RHC	55.185	5.003	0.0907
Run 4 RHC	32.586	2.883	0.0885
RHC	154.63	11.082	per run
Combined	250.49	18.848	per run

FHC Run 1 is the non-open-trigger data: $2.192 \times 10^{20} \text{ POT}$, 5 748 692 triggers (runs 4952–6748). Pseudo-data are thrown run by run at each run's exposure.

Beam direction in detector coordinates

(a) beamline geometry

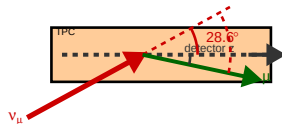
where the detector sits relative to the NuMI beam



This angle says nothing about how the TPC is oriented.

(b) inside the detector

the same ν , in TPC coordinates

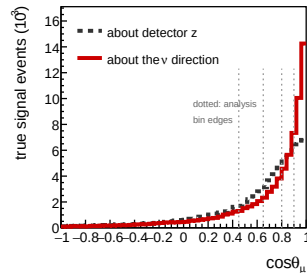


the same μ is assigned
 $\theta_\mu = 12^\circ$ from detector z
 $\theta_\mu = 40.6^\circ$ from the ν
 the generators mean the second

(c) the consequence

true CC1 π signal, FHC

64.6% of events change bin



- ▶ Off-axis angle 7.71° (134.6 mrad): property of the beamline.
- ▶ Neutrinos arrive 28.6° from detector z (mean true direction (0.462, 0.049, 0.885)).
- ▶ Generators use θ about the ν ; about z , 64.6% of events change $\cos \theta_\mu$ bin (44.6% for $\cos \theta_\pi$).

- ▶ Truth: per-event ν direction. Reco: NuMI target to reconstructed vertex (β convention of the NuMI ν_e CC1 π analysis).
- ▶ Fixed axis vs. per-event β : median 2.4 mrad, 0.42% of muons change bin.
- ▶ Momenta, $\theta_{\mu\pi}$, $W_{\pi p}$, totals and the selection do not depend on the axis.

Samples, event weights and beam-off normalisation

Sample	Composition	n
ν_μ overlay FHC	Runs 1, 2, 4 + Run 5 (rew. PPFX)	4
ν_μ overlay RHC	Runs 1, 2, 3, 4a, 4b, 4c	10
Dirt	GENIE NuMI dirt per run and mode	7
EXT (beam-off)	cosmic data per run, mode-independent	7
Detector var.	Run-4 FHC + RHC, CV + 8 knobs	18
Pseudo-data	Poisson-thrown CV, per run	

Weight $w = \text{Scale}[\text{run}] \times w_{\text{PPFX}} \times w_{\text{tune}}$, set to 1 if non-finite, negative or > 30 . The same rule in universes, covariance, cut-flow, control regions and the beam-on comparison.

Beam-off (EXT)

$$\text{EXT}_{\text{run}} = 0.98 \frac{\text{triggers}_{\text{on}}[\text{run}]}{\sum_{\text{files}} \text{triggers}_{\text{off}}} \sum_{\text{files}} N_{\text{EXT}}^{\text{sel}}$$

- ▶ Matched by run period, independent of horn polarity.
- ▶ No Run-2 beam-off: Run 1 used (both pre-CRT).
- ▶ Selected rate per 10^6 triggers: 1.75 (R1), 0.43 (R3b), 1.15 (R4), 0.52 (R5).
- ▶ At the final cut: 23 (FHC), 23 (RHC), 46 (comb.) events, 1.3–1.6% of the sample.
- ▶ No extra cosmic cut: a data-driven subtraction of ≈ 5 events' uncertainty; every variant removes more signal.

Signal definition and the pion threshold

- ▶ Generator level; used for efficiency and response, never as a reco cut.
- ▶ $|\nu_{\text{PDG}}| = 14$, CC, true vertex in FV.
- ▶ Exactly one μ and one $\pi^\pm$ (any momentum); a second $\pi^\pm$ below threshold makes the event background.
- ▶ Thresholds set by the efficiency turn-on (10 cm μ , 20 cm π track length).
- ▶ $\theta_{\mu\pi} < 2.6$ rad removes a region of $\approx 7\%$ purity (cosmic, dirt); applied at reco too.

Pion momentum reconstruction vs. true p_π

True p_π [GeV/c]	Bias	RMS	Reco frac.
0.080–0.113	+131%	73%	3.4%
0.113–0.150	+66%	56%	3.4%
0.150–0.175	+30%	51%	4.3%
0.175–0.250	−8%	31%	10.0%
0.250–0.400	−30%	21%	10.4%

The NuMI ν_e analysis uses $\text{KE}_\pi > 40$ MeV ($p_\pi > 0.113$). Below 0.175 GeV/c the few reconstructed pions are those long enough to pass, biased up; resolution 51–56%, efficiency 2.2%. No cross section is reported below 0.175 GeV/c.

Event selection

1. **NeutrinoSlice**: ≥ 1 track, software trigger
2. **InFiducialVol**: SCE-corrected reco vertex in FV
3. **Topological** score > 0.67 (EXT down $\times 28-29$)
4. **MuonCandidate**: track score > 0.8 , $d_{\text{vtx}} \leq 4$ cm, $L > 10$ cm, LLR > 0.2 , MIP BDT ≥ -0.1 ; highest muon-BDT score
5. **ContainedPion**: exactly one other track with track score ≥ 0.5 , LLR > 0.1 , $L > 20$ cm, MIP BDT > -0.1 , pion BDT > -0.1 , Bragg- $\pi \geq 0.08$, end contained
- 6-7. **μ, π in 3 planes**
8. **ShowerCut**: no primary shower, except one small shower (< 50 Y hits, > 10 cm)
9. **OpeningAngle** $\theta_{\mu\pi} < 2.6$ rad
10. **Nonprotons**: < 4 tracks with LLR > 0.1 , < 5 tracks

BDTs (TMVA) use Bragg likelihoods, LLR, track score and end points; the candidate-PID BDTs do not use track length, so they do not sculpt the momentum spectra.

Kinematics

- ▶ p_μ : range if contained (37%), MCS if exiting (63%); MCS bias -10% , resolution 30%.
- ▶ p_π : range under μ hypothesis, converted to π at fixed KE in the bin definition (lowest bin reco/true $0.888 \rightarrow 0.976$).
- ▶ Angles about $\hat{u}_\nu$; $\theta_{\mu\pi}$ from track directions only.

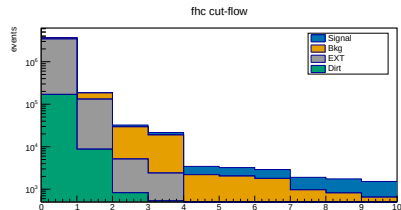
Final cut	Efficiency	Purity
FHC	17.5%	58.5%
RHC	17.6%	57.4%
Combined	17.6%	57.9%

Cut-flow at full exposure

FHC, 7.766×10^{20} POT (CV weight, per-run scaling)

Cut	Signal	Bkg	EXT	Dirt	Eff%	Pur%
None	4 753	235 281	3 256 689	170 169	100.0	0.1
InFiducialVol	3 955	52 835	122 860	8 738	83.2	2.1
Topological	2 995	24 262	4 299	827	63.0	9.2
MuonCandidate	2 689	16 484	1 870	530	56.6	12.5
ContainedPion	1 195	1 823	173	90	25.1	36.4
MuonIn3Planes	1 143	1 722	144	66	24.1	37.2
PionIn3Planes	1 062	1 549	113	37	22.3	38.5
ShowerCut	896	783	83	31	18.9	50.0
OpeningAngle	890	710	32	10	18.7	54.2
Nonprotons	833	560	23	6.6	17.5	58.5

Final cut	Signal	Bkg	EXT	Dirt	Eff%	Pur%
RHC	1 015	720	23	9.5	17.6	57.4
Combined	1 848	1 280	46	16	17.6	57.9



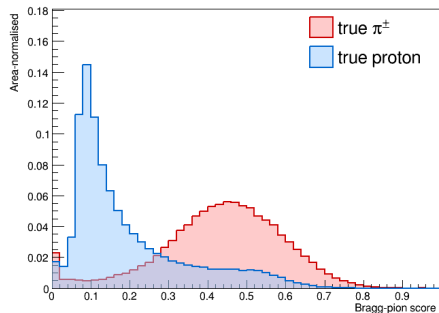
Without the opening-angle cut: purity 54.2%, EXT
64 events.

Background composition

FHC final cut, 560 neutrino-background events

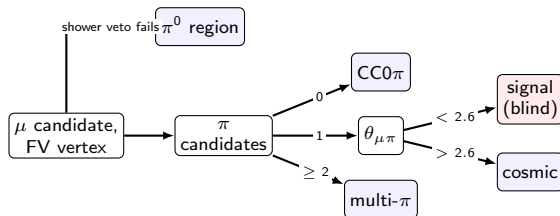
Category	Events	Frac.
CC0 π (proton or other track as π)	215	38%
CC $\geq 2\pi^\pm$, no π^0	128	23%
NC and ν_e CC	79.0	14%
CC with a π^0	51.4	9%
CC1 $\pi^\pm$ outside phase space	46.5	8%
Out of FV	41.3	7%
EXT	23	
Dirt	6.6	

- ▶ At the working point the pion candidate is a proton in $\approx 18\%$ of selected events.
- ▶ Tighter Bragg- π cuts (0.20–0.30) do not improve the FoM: $\pi : p \approx 4 : 1$ among pion candidates in signal, and a tighter cut removes both at similar rates.



Bragg- π score for true $\pi^\pm$ and protons at the final cut; threshold ≥ 0.08 .

Background control regions



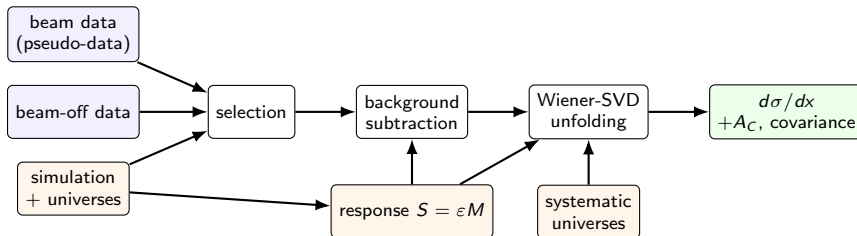
- ▶ Each region inverts one requirement; zero events shared with the signal region (checked event by event).
- ▶ CC0 π : dominant background class, $T \approx 0.03$, transfer uncertainty 31–34% (detector).
- ▶ π^0 : target CC+NC π^0 , purity 50–51%, $T \approx 0.02$.
- ▶ multi- π and cosmic: diagnostic.
- ▶ About 30% of the neutrino background has no region of its own.

Expected yields, FHC

Region	Signal	ν bkg	EXT	Dirt	Sig. frac.
CC0 π	987	9 703	1 080	342	9.2%
multi- π	27	83	5.4	0.5	24.4%
π^0	582	5 265	613	93	10.0%
cosmic	5.1	60	40	18.5	7.8%

- ▶ Studied with pseudo-data only: large regions within 1% of unity; full-stack test (with EXT and dirt thrown) 0.96–1.12.
- ▶ The comparison with beam-on data follows a frozen procedure (global χ^2 over 8 region $\times$ mode totals, pulls, shape tests, pre-declared responses), backup.
- ▶ The control regions contain unselected signal, so they will be opened after approval and before the signal region.
- ▶ Background-fit toys: pull widths 0.90–1.08.

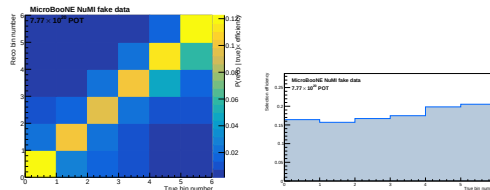
Extraction chain



$$d_i^{\text{sig}} = d_i^{\text{data}} - b_i^{\nu} - b_i^{\text{EXT}} - b_i^{\text{dirt}}$$

$$n_i = \sum_j S_{ij} N_j + b_i, \quad S_{ij} = \varepsilon_j M_{ij}$$

- S carries the efficiency, so $\hat{N}_j$ is efficiency corrected; no separate $1/\varepsilon_j$.
- Up to half of a bin comes from other true bins: direct inversion is unstable.
- $\text{conv} = N_{\text{Ar}} \Phi / 10^{-38}$: 4608 (FHC), 6223 (RHC), 10830 (comb.).



FHC p_μ : response S (left) and efficiency (right). Efficiency is 13–20% across all observables (the smallest $\theta_{\mu\pi}$ bin 9%).

Binning and the migration criterion

- Migration diagonal M_{jj} : fraction of selected signal of true bin j reconstructed in bin j .
- BNB $CC1\pi^\pm$ used 0.68; here 63% of muons are MCS (30% resolution) and p_π saturates near 0.34 GeV/ c .
- **Adopted**: every diagonal ≥ 0.50 in FHC and RHC, ≥ 70 selected signal per bin (50 proton-tagged), fewest bins reaching 90% of the largest trace of A_C .
- Trace of A_C = degrees of freedom the data determine.
- Under 0.68: p_μ 4 bins, p_π 2 regions, $\cos\theta_\mu$ 7 bins.
- The stronger model dependence admitted by 0.50 is tested by ensembles and reweighted-model pseudo-data.

Information content, FHC

Observable	bins	min. diag.	A_C row sums	tr A_C	$W_k \geq 0.5$	unc.
p_μ	6	0.57	0.45–0.98	2.09	2	42%
p_π	3	0.51	0.52–0.91	1.57	2	60%
$\cos\theta_\mu$	8	0.66	0.49–0.70	4.00	4	43%
$\cos\theta_\pi$	5	0.74	0.68–0.78	3.11	4	42%
$\theta_{\mu\pi}$	5	0.79	0.52–1.04	2.35	3	38%

Edges

p_μ (GeV/ c)	0.15, 0.3, 0.425, 0.575, 0.825, 1.25, open
p_π (GeV/ c)	0.175, 0.205, 0.26, open
$\cos\theta_\mu$	−1, 0.4, 0.625, 0.75, 0.825, 0.9, 0.95, 0.975, 1
$\cos\theta_\pi$	−1, −0.1, 0.35, 0.6, 0.8, 1
$\theta_{\mu\pi}$ (rad)	0, 0.6, 0.85, 1.3, 1.85, 2.6

Wiener-SVD and the additional-smearing matrix A_C

- ▶ Each SVD mode weighted by $W_k = \sigma_k^2 / (\sigma_k^2 + N_k)$, set by the data covariance, not tuned by hand.
- ▶ Regulariser: second derivative of the density with the actual bin widths (widths vary up to 56:1); a uniform stencil drives A_C to rank one (backup).
- ▶ The result estimates $A_C \times x_{\text{true}}$. Every prediction is smeared by A_C once:
 $\text{smeared}_i = \sum_j (A_C)_{ij} p_j / w_i$.
- ▶ A_C does not preserve normalisation: integrals differ between observables. The total is taken from the one-bin extraction ($A_C \equiv 1$).
- ▶ D'Agostini cross-check: integral flat across observables to 1% and equal to the one-bin total.

A_C row sums			
	FHC	RHC	Comb.
p_μ	0.45–0.98	0.35–0.72	0.45–0.73
p_π	0.52–0.91	0.45–0.89	0.45–0.89
$\cos \theta_\mu$	0.49–0.70	0.26–0.88	0.47–0.67
$\cos \theta_\pi$	0.68–0.78	0.70–0.84	0.70–0.83
$\theta_{\mu\pi}$	0.52–1.04	0.58–1.15	0.63–1.20

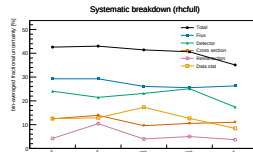
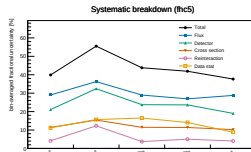
FHC p_μ : smeared/true integral 0.77–0.79 for all four generators, while their true totals span 0.82–1.32. The common factor does not create the model differences.

Released: curves (already smeared), A_C , covariance by source; external reproduction of all 96 generator curves to 4×10^{-8} .

Systematic uncertainties

Bin-averaged, 15 inclusive differential results

Source	Method	Range (%)
Flux	PPFX, 600 univ.	22.8–38.5
Detector	Run-4 var. samples	16.6–34.2
Cross section	GENIE multisim+unisims	8.4–16.6
Re-interaction	Geant4Reweight	3.5–13.3
POT		2.8–4.5
Targets		1.4–2.2
MCS scale	data-side $\pm 5\%$	0.0–10.8
MC stat		1.9–6.1
EXT stat		0.5–2.3
Data stat		6.6–19.3
Prediction total		32.6–56.7
Total		33.7–60.0



- Flux and detector dominate; data statistics is smaller in every observable.
- Combined configuration: the 600 PPFX universes are index-matched in all 14 overlays, so the FHC–RHC flux correlation is built in; detector variations pooled once per mode.
- Per-observable tables for all three configurations in backup.

The flux term: normalisation times $(1 + B/S)$

- ▶ Flux normalisation uncertainty $\approx 17\%$; flux term of the one-bin total 28–30%.
- ▶ A flux universe moves signal and neutrino background together; B and Φ move together in $(N - B)/(\varepsilon\Phi N_{Ar})$:

$$\frac{\delta\sigma}{\sigma} = \frac{\delta\Phi}{\Phi} \left(1 + \frac{B}{S}\right)$$

- ▶ FHC: $17.7\% \times (1 + 560/833) = 29.6\%$ vs. 29.9% extracted.
- ▶ RHC: $15\% \times (1 + 720/1015) = 25.6\%$ vs. 27.7%.
- ▶ Not an unfolding effect: the one-bin total has no regularisation and carries the same factor.
- ▶ Reduced only by higher purity or a control-region background constraint.

Outside the official covariance

- ▶ Beamline-geometry flux (12 unisims): 1.6–3.3%, adds $\leq 0.2\%$ to the flux row.
- ▶ EXT trigger ratios and the Run-1-for-Run-2 beam-off stand-in: a 20% uncertainty would add 0.3%.

Muon momentum scale

- ▶ MCS measures 63% of muons, not varied by the detector samples.
- ▶ Pseudo-data rethrown with MCS momenta $\times (1 \pm 0.05)$, unfolded with the nominal response; half-difference, fully correlated, in the official covariance.
- ▶ FHC p_μ : 5–8% ($\leq 0.19\sigma$) in the five lower bins, 17% (0.56σ) in the open top bin.
- ▶ Zero for observables that do not use p_μ . The $\pm 5\%$ size is assumed.

Validation: from the implementation to the model

Step	Question	Result
Identity closure	$UR = A_C$ exactly?	All 59 extractions; worst 6.1×10^{-5} relative (FP rounding)
Nominal closure	Truth of CV pseudo-data returned?	Inclusive $\chi^2 \leq 2.39/8$, integral ratios 0.946–1.047 (mean 0.996); proton-tagged $\chi^2 \leq 6.01/4$
Ensembles	Statistical uncertainties sized correctly?	7 inclusive + 10 proton-tagged, 100 members each: pull widths 0.95–1.07 / 0.94–1.08; 68% coverage 63–73%
Rewighted models	Follows a model reweighting can reach?	GENIE multisim u545 and $\Delta \rightarrow N\pi$ angular: largest bias 1.18σ (one sparse FHC $\cos\theta_\mu$ bin); RHC $\leq 0.83\sigma$
Independent generator	Follows a model reweighting cannot reach?	Not made: the NuWro overlay has $1.76\times$ the GENIE rate vs. 1.6 expected, provenance untraced
Control regions	Background model supported by data?	Pseudo-data only; frozen procedure for beam-on

Closure χ^2 with the full covariance clusters near $p = 1$: systematic terms are common to pseudo-data and reference. It checks the implementation; ensembles test the sizing.

Ensemble means sit 0.4–1.1% below the reference (pull means -0.05 to -0.11 , RHC total -0.21), including the unregularised totals; not corrected.

Ensembles and reweighted-model pseudo-data

Ensembles (statistical covariance)

Extraction	mean	width	68/95%
p_μ , FHC	-0.11	1.07	64/94
$\cos \theta_\mu$, FHC	-0.06	0.97	72/95
$\cos \theta_\mu$, RHC	-0.05	1.05	63/94
p_π , comb.	-0.11	0.95	73/96
total, FHC	-0.10	0.97	67/95
total, RHC	-0.21	0.99	64/94
$\cos \theta_\pi \times \cos \theta_\mu$, comb.		1.03	67/94

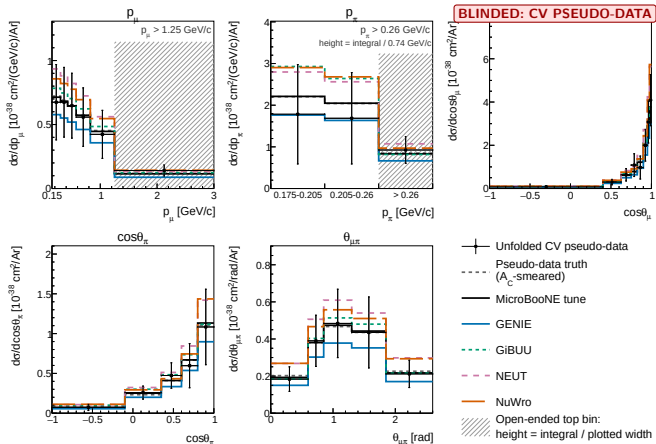
Each member rethrown run by run, universes rebuilt; pull against CV truth smeared by the member's own A_C . Chosen to span conditioning ($\cos \theta_\mu$ RHC row sum 0.26) and the unregularised totals.

Reweighted-model pseudo-data, nominal response

Obs.	Model	FHC		RHC	
		ratio	bias	ratio	bias
p_μ	u545	0.920	-0.47σ	0.783	-0.83σ
	Δ ang.	0.992	$+0.83\sigma$	1.027	$+0.56\sigma$
$\cos \theta_\mu$	u545	0.942	-0.87σ	1.059	$+0.78\sigma$
	Δ ang.	0.816	-1.18σ	1.045	-0.47σ
$\cos \theta_\pi$	u545	0.980	$+0.40\sigma$	0.852	-0.64σ
	Δ ang.	0.944	-0.59σ	0.987	$+0.20\sigma$
p_π	u545	0.750	-0.70σ	0.852	-0.36σ
	Δ ang.	1.084	$+0.34\sigma$	1.158	$+0.55\sigma$

u545 changes shapes by up to 29% at 4% rate change. Bins that do not follow the alternative truth are sparse ones (near-threshold p_π , open top p_μ , $\cos \theta_\mu$ bins 1 and 3), with uncertainties of 29–114%. The test should be repeated if the uncertainties shrink substantially.

Differential cross sections, FHC

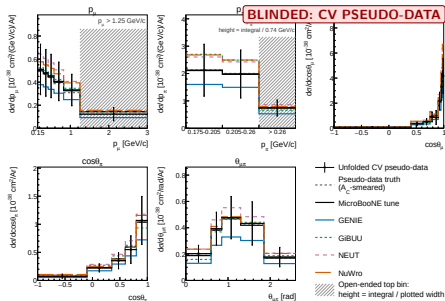


	σ_{int}	Flux	Det.	Closure
p_μ	0.814	30.0%	21.2%	0.82/6
p_π	0.830	38.5%	34.2%	1.01/3
$\cos\theta_\mu$	0.713	26.6%	25.4%	2.39/8
$\cos\theta_\pi$	0.668	27.1%	23.2%	1.21/5
$\theta_{\mu\pi}$	0.796	28.8%	19.9%	0.39/5

- Points: unfolded pseudo-data, total uncertainty. All curves A_C -smeared.
- σ_{int} in $10^{-38} \text{ cm}^2/\text{Ar}$; differs between observables through A_C .
- Open top bins hatched: height = integral / plotted width.

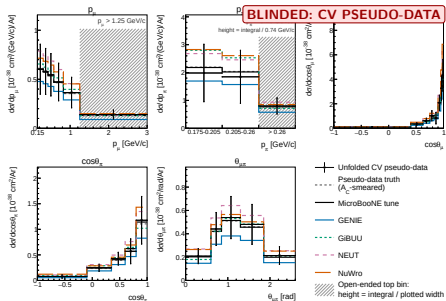
Differential cross sections, RHC and combined

RHC



	p_μ	p_π	$\cos \theta_\mu$	$\cos \theta_\pi$	$\theta_{\mu\pi}$
σ_{int}	0.644	0.727	0.694	0.615	0.831
Syst. total	42.0%	41.1%	36.9%	38.5%	32.6%
Closure χ^2	0.44/6	0.01/3	0.98/8	0.49/5	0.18/5

Combined



	p_μ	p_π	$\cos \theta_\mu$	$\cos \theta_\pi$	$\theta_{\mu\pi}$
σ_{int}	0.742	0.762	0.744	0.649	0.860
Syst. total	40.1%	45.9%	33.1%	37.5%	33.6%
Closure χ^2	0.08/6	0.21/3	1.29/8	0.58/5	0.21/5

Generator predictions averaged over the flux of each configuration; combined: fluence-weighted average (0.425 FHC, 0.575 RHC).

Pion momentum in three regions and model comparisons

p_π as bin integrals ($10^{-38} \text{ cm}^2/\text{Ar}$)

Region	FHC	RHC	Comb.
0.175–0.205	0.048 ± 0.036	0.064 ± 0.029	0.059 ± 0.032
0.205–0.26	0.086 ± 0.061	0.110 ± 0.049	0.102 ± 0.053
> 0.26	0.695 ± 0.236	0.553 ± 0.222	0.601 ± 0.216

- ▶ Range estimator saturates near 0.34 GeV/c: pions interact before stopping.
- ▶ Diagonals 91/51/52%; the two near-threshold regions are measured as a pair (A_C mixes them strongly, trace 1.57).
- ▶ Upper region is open: compare bin integrals after A_C , not densities.

χ^2/ndf against A_C -smeared predictions, full covariance

		truth	tune	GENIE	GiBUU	NEUT	NuWro
FHC	p_μ	0.82	1.29	2.24	1.27	2.27	1.35
	p_π	1.01	1.20	1.56	4.13	1.83	2.26
	$\cos \theta_\mu$	2.39	1.75	2.37	2.57	3.93	4.88
	$\cos \theta_\pi$	1.21	1.21	1.92	2.56	2.26	3.09
	$\theta_{\mu\pi}$	0.39	0.47	0.82	0.67	2.71	3.16
RHC	$\cos \theta_\mu$	0.98	1.27	1.84	0.71	6.36	10.39
comb	$\cos \theta_\mu$	1.29	1.17	2.17	1.30	5.24	9.52

ndf = 6, 3, 8, 5, 5. Full table in backup. With pseudo-data from the tune, these test the procedure; the spread shows the model differences the measurement is expected to resolve. A_C is applied once; applying it twice created a spurious NEUT/NuWro separation (fixed).

Total cross section: one-bin extraction

$$\sigma = \frac{N_{\text{sel}} - N_{\text{bkg}}}{\epsilon \Phi N_{\text{Ar}}}, \quad A_C \equiv 1$$

- ▶ One true and one reco bin over $\cos \theta_\mu \in [-1, 1]$; Wiener filter off; every term evaluated universe by universe.
- ▶ Closed-form evaluation reproduces the CV one-bin extraction to four digits (1.0315, 0.9285, 0.9723).
- ▶ Flux 28–30%, detector 20–22%, data statistics 3–5%.
- ▶ RHC/FHC: tune 0.900, extracted 0.901.
- ▶ The one-bin total lies above the A_C -suppressed σ_{int} and agrees with D'Agostini.

Generators vs. one-bin total ($10^{-38} \text{ cm}^2/\text{Ar}$)

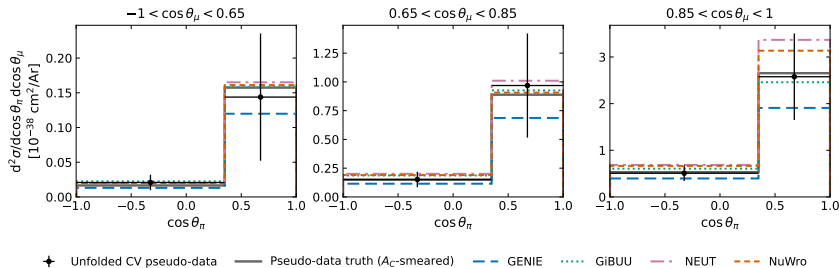
	FHC	RHC	Comb.
Measured	1.03 ± 0.40	0.93 ± 0.35	0.98 ± 0.37
GENIE	0.82	0.67	0.73
GiBUU	1.13	0.94	1.02
NEUT	1.32	1.10	1.20
NuWro	1.25	1.04	1.13

All within 0.8σ ; generator spread a factor 1.6.

Config	N_{sel}	N_{bkg}	ϵ	σ	stat	syst	total	truth	σ/truth	$\chi^2/1$
FHC	1423	590	17.5%	1.034 ± 0.396	4.5%	38.0%	38.3%	1.044	0.991	0.001
RHC	1768	752	17.6%	0.932 ± 0.350	4.1%	37.4%	37.6%	0.941	0.990	0.001
Combined	3191	1342	17.5%	0.975 ± 0.366	3.0%	37.5%	37.6%	0.985	0.990	0.001

Two-dimensional $d^2\sigma/d\cos\theta_\pi d\cos\theta_\mu$ (combined, secondary)

$\cos\theta_\pi \times \cos\theta_\mu$, FHC+RHC combined (CV pseudo-data; signal region b **BLINDED: CV PSEUDO-DATA**)



- $\cos\theta_\pi$: -1, 0.35, 1 inside $\cos\theta_\mu$: -1, 0.65, 0.85, 1 (6 bins).
- Combining modes raises the A_C conditioning from 0.245 (FHC), 0.315 (RHC) to 0.564; single-mode extractions not reported.
- Closure $\chi^2 = 0.88/6$, cell ratios 0.91–1.26; per-bin uncertainty 33–64%.

- Ensemble: pull width 1.03, coverage 67/94%.
- Sparsest cell 11% low on average (0.58 stat. s.d., 0.17 of its total), not corrected.
- Reweighted models: every cell within 0.61σ .

Proton-tagged subsample: selection and observables

Signal: inclusive signal with ≥ 1 true proton above 0.3 GeV/c; leading proton for truth observables.

Cut 11: most proton-like remaining track (LLR < 0.05), contained, $p_p \geq 0.3$ GeV/c (range KE).

	FHC	RHC	Comb.
Efficiency	11.5%	11.4%	11.5%
Purity	52%	50%	51%

- ▶ Rejects 67% of the neutrino background for 32% signal loss; EXT down $\times 3.3$ – 3.6 .
- ▶ Candidate is the leading true proton in 81.7%, some proton in 87.4%, not a proton in 12.6%.
- ▶ Wrong candidates dominate the θ_p tails (74% of residuals beyond 0.3 rad).

Observables (about the ν direction)

- ▶ p_p (3), θ_p (4), $\theta_{\pi p}$ (4)
- ▶ W_{had} (2 regions, split 1.18 GeV/c²)
- ▶ TKI of μ vs. ($\pi + p$): δp_T (2), $\delta \alpha_T$ (2), $\delta \phi_T$ (3), p_n (2)
- ▶ p_μ , p_π (2), $\cos \theta_\mu$, $\cos \theta_\pi$ (4), $\theta_{\mu\pi}$ again on this phase space

Binnings: 0.50 criterion, ≥ 50 selected FHC signal per bin.

Status

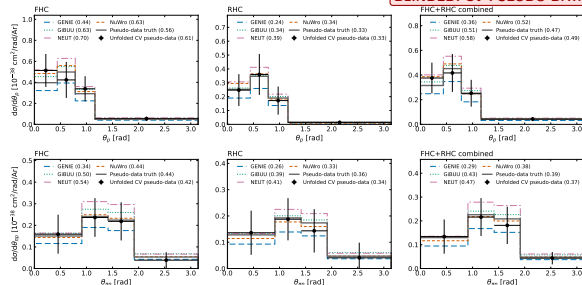
- ▶ One-bin total, $d\sigma/d\theta_p$, $d\sigma/d\theta_{\pi p}$ (3 configs), $d^2\sigma/d\theta_p d\delta p_T$ (combined).
- ▶ W_{had} , TKI, p_p and the μ/π observables are released as validation-only products.
- ▶ No differential $W_{\pi p}$ result.

Detector term 9.9–74.5% (median 30.9%).

Proton-tagged results

Proton-tagged CC $1\mu\text{m}^2$: new observables, CV pseudo-data (signal region blind); curves a

BLINDED: CV PSEUDO-DATA



θ_p (top) and $\theta_{\pi p}$ (bottom), FHC, RHC, combined.

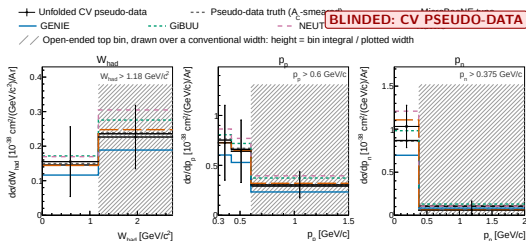
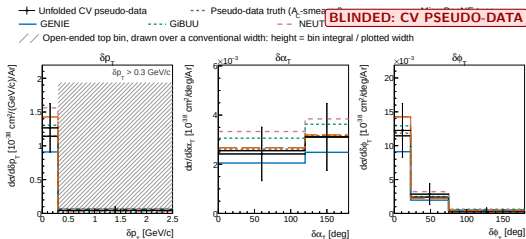
One-bin total ($10^{-38} \text{ cm}^2/\text{Ar}$)

	σ	total	truth	σ/truth
FHC	0.608 ± 0.256	42.0%	0.639	0.952
RHC	0.531 ± 0.251	47.2%	0.532	0.998
Comb.	0.564 ± 0.239	42.5%	0.578	0.976

FHC generators: GENIE 0.50, NuWro 0.65, GiBUU 0.74, NEUT 0.78; tune 0.63. RHC/FHC 0.873 (tune 0.878).

- ▶ $\theta_{\pi p}$: trace 2.1–2.2, three components with $S/N > 1$ (FHC, RHC); last bin 94–141%.
- ▶ θ_p : trace 2.1–2.3; largest per-bin 41% (FHC), 84% (RHC).
- ▶ Ensembles: widths 0.94–1.03; reweighted models $\leq 0.79\sigma$.
- ▶ Generators differ mainly in normalisation.

Proton-tagged: TKI, W_{had} and model sensitivity



FHC: δp_T , $\delta \alpha_T$, $\delta \phi_T$ (top); W_{had} , p_p , p_n (bottom).

- Closure: $\chi^2 \leq 6.01/4$ (RHC $\cos \theta_\pi$), integral ratios 0.93–1.18.
- All generators compatible with the pseudo-data; largest χ^2 4.1/2 (GENIE, comb. p_n).
- Under GENIE u545 the low δp_T and p_n bins return 35% low (-1.14σ , -1.16σ) while their truth moves 1%: the change enters through sample composition and proton-tag efficiency.
- W_{had} and p_n use a residual-mass prescription derived for QE kinematics, not validated for pion production.
- MCS scale term recomputed for W_{had} , δp_T , $\delta \alpha_T$, p_n : $\leq 0.37\sigma$.

Why no $W_{\pi p}$: six bins have diagonals down to 2%; the best two-bin split passes 0.50 by 0.003 with a second filter factor 0.27–0.29 (normalisation only).

Limitations

- ▶ **Blind signal region:** results demonstrate the extraction and its uncertainties, not the cross section. At 57–59% purity the background model enters strongly; closure cannot test it.
- ▶ **Pion momentum:** saturation near 0.34 GeV/c; golden pion, MCS, calorimetry, regression, daughter recovery and the proton tag all tried. A truth-level golden selection does no better than the classifier, so the limit is physical.
- ▶ **Model dependence:** reweighting bounds the bias at 1.18σ ; no independent-generator closure. Efficiency model dependence matters most for the one-bin total and the open $p_\pi > 0.26$ region.
- ▶ **Muon momentum scale:** $\pm 5\%$ assumed, not yet compared with MCS vs. range on contained muons in data.
- ▶ **Proton-tagged:** no control region constrains the 12.6% pion→proton mis-assignment (an inverted proton-PID region is not defined); no covariance term for the TKI model sensitivity.
- ▶ **Background control regions:** studied with pseudo-data; beam-on comparison follows the frozen procedure.

Summary

- ▶ $\nu_\mu + \bar{\nu}_\mu$ CC $\pi^\pm$ on argon with the full NuMI exposure: FHC 7.766, RHC 11.082, combined 18.848×10^{20} POT. Blind; all results from CV pseudo-data.
- ▶ Selection: 17.5% efficiency at 57–59% purity; angles about the per-event ν direction, 28.6° from the detector axis.
- ▶ Primary results: $d\sigma/dp_\mu$, $d\sigma/d\cos\theta_\mu$, $d\sigma/d\cos\theta_\pi$, $d\sigma/d\theta_{\mu\pi}$, p_π in three regions, and the one-bin total, in three configurations. Secondary: combined $\cos\theta_\pi \times \cos\theta_\mu$.
- ▶ Total: 38% uncertainty (flux 28–30% = $17\% \times (1 + B/S)$, detector 20–22%). Differential: 33–46% bin-averaged without data statistics.
- ▶ Every result released with A_C and covariance by source; binnings meet the 0.50 migration criterion.
- ▶ Validation: closure mean ratio 0.996; ensemble pull widths 0.95–1.07; reweighted-model bias $\leq 1.18\sigma$; $UR = A_C$ to 6.1×10^{-5} ; independent rebuild reproduces all 51 extractions to 10^{-9} .
- ▶ Proton-tagged: one-bin total, θ_p , $\theta_{\pi p}$ and combined $\theta_p \times \delta p_T$; 11.5% efficiency, 50–52% purity.
- ▶ Outstanding before unblinding: beam-on control-region comparison, independent-generator closure.

Backup

Technical supplement v1.6

Final configuration · evidence for the choices · rejected alternatives

Backup: final binnings and the binning rule

Rule (binning review, 2026-09-26)

1. Maximin partition for every bin count up to 12 (largest worst diagonal over FHC and RHC; dynamic programming, `maximin_binning.py`).
2. Event floor: ≥ 70 selected FHC signal (inclusive), ≥ 50 (proton-tagged).
3. Every diagonal ≥ 0.50 in FHC and RHC on the analysis sample.
4. Extract candidates in full, three configurations.
5. Wiener filter factors; trace of A_C .
6. Fewest bins reaching 90% of the largest trace (FHC/RHC average); ties to larger worst diagonal.
7. Check closure and per-bin uncertainties.
8. Freeze.

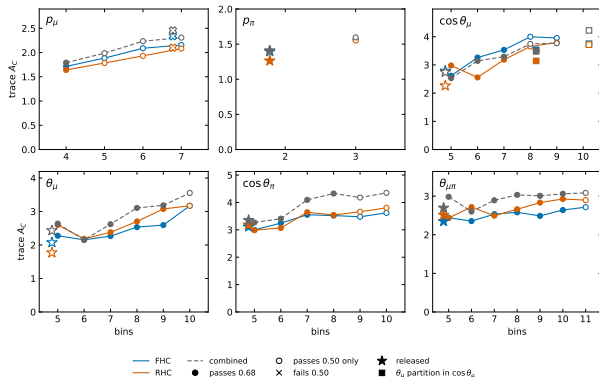
Exceptions: $\cos \theta_\pi$ and $\theta_{\mu\pi}$ keep 5 bins; $\delta\alpha_T$ keeps 120° ; proton-tagged p_π keeps 2 regions. 153 extractions in the review.

Proton-tagged edges

Obs.	bins	edges
p_π	2	0.175, 0.205, open
$\cos \theta_\pi$	4	-1, -0.1, 0.35, 0.75, 1
W_{had}	2	0, 1.18, open
δp_T	2	0, 0.3, open
$\delta\phi_T$	3	0, 22.5, 75, 180°
$\delta\alpha_T$	2	0, 120, 180°
p_n	2	0, 0.375, open
p_p	3	0.3, 0.42, 0.6, open
θ_p	4	0, 0.44, 0.785, 1.162, π
$\theta_{\pi p}$	4	0, 0.911, 1.414, 1.916, π

p_μ , $\cos \theta_\mu$, $\theta_{\mu\pi}$ use the inclusive edges. Open bins are drawn with a conventional width; the bin integral is the quantity.

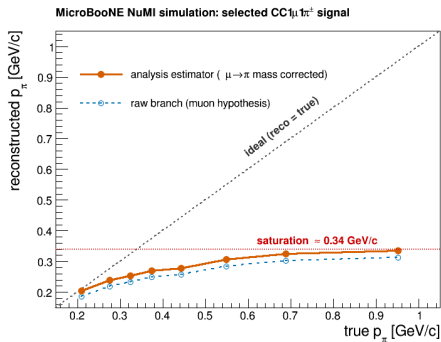
Backup: binning review, 0.68 vs. 0.50



Trace of A_C vs. number of bins, inclusive. Filled: passes 0.68; open: 0.50 only; cross: neither.

- Where the criteria differ, 0.50 adds 0.2–0.6 degrees of freedom per beam: p_μ 6 vs. 4, p_π 3 vs. 2, $\cos\theta_\mu$ 8 vs. 7, $\delta\phi_T$ 3 vs. 2, $\theta_{\pi p}$ 4 vs. 3.
- 0.68 would drop p_μ from three well-measured components to two.
- Released 7-bin p_μ failed both (diagonal down to 0.33); 6-bin W_{had} failed both.
- $\cos\theta_\pi$ at 7 bins would add 0.45 d.o.f.; kept at 5.
- $\theta_{\mu\pi}$ carries ≈ 2.5 d.o.f. however it is binned.
- Conditioning s_{min}/s_1 replaced by the trace: it is set by the weakest component alone.

Backup: pion momentum saturation



true p_π	N	$\langle p^{\text{true}} \rangle$	$\langle p^{\text{corr}} \rangle$	corr./true
0.175–0.250	528	0.210	0.205	0.976
0.250–0.300	270	0.276	0.238	0.863
0.300–0.350	254	0.325	0.253	0.779
0.350–0.400	265	0.374	0.270	0.721
0.400–0.500	403	0.443	0.279	0.629
0.500–0.600	252	0.548	0.307	0.560
0.600–0.800	272	0.688	0.326	0.473
0.800–1.200	164	0.951	0.336	0.353

- ▶ $\mu \rightarrow \pi$ mass swap fixes the hypothesis bias; nothing fixes the hadronic truncation.
- ▶ Five-bin diagonals 94/37/23/10/6%: every true bin piles into reco bin 1.
- ▶ Two regions: 91/76% (passes 0.68); three regions: 91/51/52% (passes 0.50).
- ▶ First edge 0.205: narrowest interval with BNB precedent (30 MeV/c); best in every edge scan.
- ▶ Proton tag does not change the pion response (88.9/74.2% vs. 91.0/75.5%).

Backup: pion momentum estimators and the golden-pion classifier

Golden-pion BDT, held-out Run 2, track sample

cut	eff.	golden	b2	b3	b4	b5
none	100%	43.1%	23.7	13.2	6.4	5.5
> 0.00	44.8%	64.9%	33.1	24.5	15.4	9.1
> +0.15	15.2%	84.5%	46.9	42.2	36.1	19.5
truth golden	43.1%	100%	36.1	24.1	14.2	8.9

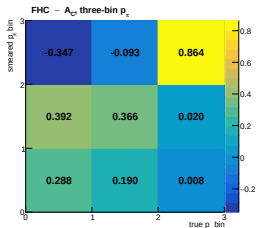
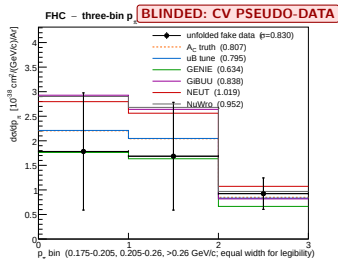
ROC 0.758; reproduces the BNB benchmark (36 $\rightarrow$ 68% golden). At matched efficiency the classifier equals perfect truth selection bin by bin: no better classifier can repair the response. BNB meets 0.68 because its resolved bins lie below 0.175 GeV/c.

Estimators without a magnetic field

estimator	reco/true (top)	RMS	
range	0.319	0.16	saturates
calorimetric	0.383	0.31	truncated, noisier
MCS	1.011	2.1	unusable on short tracks
golden selection	0.327	0.18	no gain > 0.8

A pion that interacts before stopping destroys the information a non-magnetised detector uses. Regression removes the $W_{\pi p}$ bias at the cost of model dependence. Not tried: TKI constraint on p_T^π in the proton-tagged sample (limited by Fermi motion, ≈ 250 MeV/c).

Backup: three-region p_π result and its A_C



$d\sigma/dp_\pi$ [$10^{-38} \text{ cm}^2/(\text{GeV}/c)/\text{Ar}$], all curves A_C -smeared

	FHC			RHC			Comb.		
	b1	b2	b3	b1	b2	b3	b1	b2	b3
unfolded	1.62	1.57	0.94	2.14	2.00	0.75	1.97	1.85	0.81
unc.	1.21	1.11	0.32	0.96	0.89	0.30	1.05	0.97	0.29
A_C truth	2.39	2.21	0.85	2.10	1.97	0.77	2.24	2.07	0.78
GENIE	1.91	1.77	0.67	1.57	1.47	0.53	1.73	1.60	0.58
GiBUU	3.29	2.97	0.83	2.64	2.44	0.67	2.91	2.62	0.72
NEUT	3.09	2.83	1.08	2.56	2.39	0.88	2.77	2.53	0.94
NuWro	3.16	2.91	0.97	2.63	2.45	0.80	2.88	2.66	0.85
χ^2	1.01/3			0.01/3			0.21/3		

FHC $A_C = \begin{pmatrix} 0.40 & 0.16 & 0.01 \\ 0.58 & 0.32 & 0.02 \\ -0.22 & -0.11 & 0.85 \end{pmatrix}$: the near-threshold regions are mixed and carry one component together (trace 1.57). Negative entries are the regularisation borrowing between regions. The five-bin extraction also closes ($\chi^2 \approx 2/5$): closure cannot expose a response without shape information; the migration criterion can.

Backup: hadronic mass $W_{\pi p}$ is not reported

	$W_{\pi p}$	W_{had}
residual RMS [GeV/ c^2]	0.142	0.283
half-width (16–84%)	0.101	0.177
bias	−0.111	−0.032
narrowest six-bin bin	0.040	0.100

- ▶ Six-bin $W_{\pi p}$ diagonals 88/22/12/12/10/2%: bins 2–3× narrower than the resolution.
- ▶ $W_{\pi p}$ inherits the p_π saturation.
- ▶ Low generator χ^2 at six bins reflected a binning that cannot resolve the models, not agreement.

Two bins fail on statistics

two-bin scheme	A_C	s_2/s_1
$W_{\pi p}, N_1 = 137$	$\begin{pmatrix} 0.018 & -0.410 \\ 0.034 & 0.892 \end{pmatrix}$	0.031
$p_\pi \text{ 1p}, N_1 = 469$	$\begin{pmatrix} 0.498 & -0.089 \\ 0.013 & 0.854 \end{pmatrix}$	0.577
$p_\pi \text{ incl.}, N_1 = 1056$	$\begin{pmatrix} 0.594 & -0.382 \\ 0.011 & 0.891 \end{pmatrix}$	0.528

- ▶ Narrow enough to pass the criterion leaves bin 1 too sparse; wide enough to unfold (1.19, 468 events) fails at 56%.
- ▶ No two- or three-bin scheme between 1.12 and 1.70 satisfies both.
- ▶ W_{had} in two regions (0.71/0.72) is released as validation-only.

Backup: bin widths in the regularisation operator

- ▶ The $(1, -2, 1)$ stencil is a second derivative only on a uniform grid. With width ratios up to 14:1 its near-null direction is the constant vector.
- ▶ The Wiener filter then keeps that direction alone: rank-1 A_C , normalisation passed, shape lost.
- ▶ FHC θ_μ (pre-review binning): data, truth, tune and all generators share one shape to 4 decimals; inputs $s_2/s_1 = 5 \times 10^{-2}$, outputs 2.8×10^{-6} .
- ▶ Adopted: density differenced on the actual grid, one-sided first derivative at the ends (Dirichlet rejected: $\cos \theta_\mu$ peaks at its edge).

A_C conditioning s_2/s_1 , uniform $\rightarrow$ width-aware

	ratio	FHC	RHC	COMB
p_μ	6.3	0.34 $\rightarrow$ 0.90	0.31 $\rightarrow$ 0.94	0.19 $\rightarrow$ 0.68
$\cos \theta_\mu$	14.5	0.46 $\rightarrow$ 0.57	0.45 $\rightarrow$ 0.55	0.42 $\rightarrow$ 0.39
$\cos \theta_\pi$	3.6	0.67 $\rightarrow$ 0.71	0.55 $\rightarrow$ 0.79	0.55 $\rightarrow$ 0.74
$\theta_{\mu\pi}$	3.0	0.42 $\rightarrow$ 0.85	0.47 $\rightarrow$ 0.65	0.42 $\rightarrow$ 0.46
θ_μ	10.4	0.015 $\rightarrow$ 0.51	0.22 $\rightarrow$ 0.62	0.045 $\rightarrow$ 0.38
proton-tagged, combined				
$W_{\pi p}$				0.022 $\rightarrow$ 0.60
$\delta\phi_T$				0.083 $\rightarrow$ 0.85
δp_T				0.25 $\rightarrow$ 0.93

θ_μ is not reported: the same partition retains up to one d.o.f. more as $\cos \theta_\mu$, since the regulariser acts on the density per unit of the variable.

Backup: D'Agostini cross-check and identity closure

Integrated σ averaged over the five observables

	WSVD	D'Ag. (4 it.)	ratio	1→4 it.
FHC	0.764	1.031	1.35	$\leq 0.4\%$
RHC	0.702	0.934	1.33	$\leq 0.2\%$
Comb.	0.751	0.975	1.30	$\leq 0.1\%$

- ▶ D'Agostini flat across observables to 1.0/1.1/0.5% and equal to the one-bin totals (1.034, 0.932, 0.975); WSVD integrals spread 21–31%.
- ▶ So the spread of σ_{int} is the A_C smearing.
- ▶ With pseudo-data equal to its prior, D'Agostini returns the prior: its stability does not measure its bias for a different truth.
- ▶ Also run: bin-by-bin, Tikhonov, RooUnfold Bayes/SVD/Invert/BinByBin.

Identity closure

$$\hat{x} = Ur, \quad A_C = UR \Rightarrow \|UR - A_C\|_\infty \stackrel{!}{=} 0$$

- ▶ Independent of data; catches unfolding and A_C built from different responses (a fixed-factor bias invisible to closure χ^2).
- ▶ Runs on every extraction ([IDENTITY]); all 59 pass.
- ▶ Worst 6.1×10^{-5} (FHC $\cos \theta_\mu$, width ratio 56:1); only the two $\cos \theta_\mu$ extractions exceed 4×10^{-6} . Floating-point rounding of the width-aware operator.

Closure χ^2 interpretation: median $p = 0.98$ over 51 extractions. Statistical-only covariance raises χ^2/ndf from 0.02–0.2 to 0.5–0.8.

Backup: systematic breakdown by configuration

Bin-averaged fractional uncertainty (%), official covariance

Source	FHC					RHC					Combined				
	p_μ	p_π	$c\theta_\mu$	$c\theta_\pi$	$\theta_{\mu\pi}$	p_μ	p_π	$c\theta_\mu$	$c\theta_\pi$	$\theta_{\mu\pi}$	p_μ	p_π	$c\theta_\mu$	$c\theta_\pi$	$\theta_{\mu\pi}$
Prediction total	40.6	56.7	39.6	39.0	37.1	42.0	41.1	36.9	38.5	32.6	40.1	45.9	33.1	37.5	33.6
Cross section	11.9	16.6	10.9	11.4	10.6	12.6	13.7	9.7	10.8	10.7	11.7	14.9	8.4	10.7	10.1
Flux	30.0	38.5	26.6	27.1	28.8	29.4	27.7	24.6	25.2	25.2	27.9	31.0	22.8	25.7	25.8
Detector	21.2	34.2	25.4	23.2	19.9	23.8	24.0	23.5	25.5	16.6	23.7	26.9	21.2	23.0	18.0
Re-interaction	4.2	13.3	3.5	5.1	4.1	4.4	10.6	4.3	5.1	3.7	4.0	12.6	3.6	5.1	3.5
MCS scale	8.5	0	0	0	0	10.8	0	0	0	0	8.8	0	0	0	0
MC stat	3.8	6.1	5.2	4.2	2.8	3.7	3.7	5.3	3.6	2.4	2.8	3.3	3.4	2.9	1.9
EXT stat	1.1	2.3	1.7	1.2	1.0	1.6	1.1	1.2	1.0	0.6	1.1	1.1	0.9	0.8	0.5
Data stat	12.1	19.3	16.6	13.6	8.9	13.2	12.7	18.4	12.3	8.4	9.5	11.0	11.5	9.7	6.6
POT + targets	3.8	5.0	3.5	3.4	3.6	4.0	3.9	3.4	3.5	3.5	3.7	4.2	3.1	3.5	3.4
Total	42.4	60.0	43.1	41.5	38.3	44.2	43.1	41.4	40.5	33.7	41.4	47.3	35.2	38.8	34.2

Rows are bin averages of per-bin fractions, so they do not add in quadrature to the totals. Reconstructed-level flux term 16.4–17.7%, flat across observables. The detector term uses the native Run-4 FHC and RHC variation samples; the combined term is pooled once to the data exposure (scaling each set to the full exposure would double it and push the combined truth ratio to 0.91).

Backup: model χ^2 , all configurations

		truth	tune	GENIE	GiBUU	NEUT	NuWro
FHC	$p_\mu / 6$	0.82	1.29	2.24	1.27	2.27	1.35
	$p_\pi / 3$	1.01	1.20	1.56	4.13	1.83	2.26
	$\cos \theta_\mu / 8$	2.39	1.75	2.37	2.57	3.93	4.88
	$\cos \theta_\pi / 5$	1.21	1.21	1.92	2.56	2.26	3.09
	$\theta_{\mu\pi} / 5$	0.39	0.47	0.82	0.67	2.71	3.16
RHC	p_μ	0.44	0.76	0.83	0.57	0.88	1.24
	p_π	0.01	0.01	0.58	0.71	0.26	0.27
	$\cos \theta_\mu$	0.98	1.27	1.84	0.71	6.36	10.39
	$\cos \theta_\pi$	0.49	0.43	1.27	2.86	1.79	1.08
	$\theta_{\mu\pi}$	0.19	0.34	2.52	3.54	0.46	1.08
comb	p_μ	0.08	0.22	1.15	0.24	0.85	0.62
	p_π	0.21	0.26	0.89	2.66	0.70	1.19
	$\cos \theta_\mu$	1.29	1.17	2.17	1.30	5.24	9.52
	$\cos \theta_\pi$	0.58	0.52	1.65	2.70	1.63	2.06
	$\theta_{\mu\pi}$	0.21	0.38	1.69	2.21	1.93	2.88

- Unfolded pseudo-data vs. each A_C -smeared prediction, full official covariance.
- Pseudo-data are thrown from the tune: the comparison tests the procedure, not the generators.
- The prescription applied to the true predictions reproduces the smeared curves to 4×10^{-8} .
- Proton-tagged: largest χ^2 4.1/2 (GENIE, combined p_n), all others < 2.6 .

Backup: ensembles, full covariance and proton-tagged

Inclusive, full framework covariance

	width	68/95%
p_μ FHC	0.32	99.5/100
$\cos \theta_\mu$ FHC	0.37	98.5/100
$\cos \theta_\mu$ RHC	0.46	96.1/100
p_π comb.	0.22	100/100
total FHC	0.11	100/100
total RHC	0.11	100/100

Ensembles vary only data statistics; flux, detector and cross-section terms are the same in every member. Throw unbiased (mean selected 1578.7 vs. 1580.6). Truncated throw-file reads found and rebuilt; throws now created once (`throw_guard.h`); entry counts checked for every member.

Proton-tagged, 100 members each

Extraction	cfg	mean	statistical width	68%	95%	full width	integral offset
W_{had}	FHC	-0.05	0.94	72	97	0.26	-0.4%
δp_T	FHC	-0.04	1.08	62	92	0.40	-0.6%
p_n	comb	-0.04	0.99	64	96	0.31	-0.3%
θ_p	FHC	-0.09	1.01	68	94	0.39	-1.0%
	RHC	-0.04	0.97	70	96	0.37	-1.3%
	comb	-0.12	1.03	65	95	0.32	-1.2%
$\theta_{\pi p}$	FHC	-0.02	1.00	67	94	0.40	-0.2%
	RHC	-0.06	0.94	72	96	0.26	-0.7%
	comb	-0.01	1.02	65	94	0.35	+0.1%
$\theta_p \times \delta p_T$	comb	-0.05	1.04	67	94	0.40	-0.6%

Backup: proton-tagged reweighted-model pseudo-data (FHC)

Observable	Variation	ratio	max dev.	bias (bin)
W_{had}	u545	0.83	29%	-0.40σ (1)
	Δ ang.	1.23	58%	$+0.89\sigma$ (1)
δp_T	u545	0.66	35%	-1.14σ (1)
	Δ ang.	1.04	4%	$+0.14\sigma$ (1)
$\delta\alpha_T$	u545	0.85	24%	-0.57σ (1)
$\delta\phi_T$	Δ ang.	1.22	51%	$+0.85\sigma$ (2)
p_n	u545	1.01	34%	-1.16σ (1)
p_p	Δ ang.	1.08	17%	$+0.37\sigma$ (1)
θ_p	u545	0.82	28%	-0.79σ (2)
	Δ ang.	1.12	29%	$+0.75\sigma$ (3)
$\theta_{\pi p}$	Δ ang.	1.04	38%	$+0.67\sigma$ (1)
$\theta_p \times \delta p_T$ (comb.)	u545	0.94	26%	-0.68σ (2)
	Δ ang.	0.95	61%	-0.74σ (3)

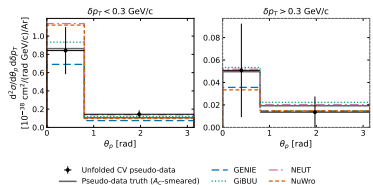
- ▶ Selected rows; full table in the proton-tagged note.
- ▶ Low-imbalance δp_T and p_n bins, closest to QE kinematics, return 35% below the alternative truth while that truth moved 1%.
- ▶ The shift enters through the response and background: the varied model changes the sample composition and the proton-tagging efficiency.
- ▶ Covered at about 1σ ; no covariance term for this model dependence.
- ▶ θ_p integral varies 12% throw to throw, so a single throw cannot separate a model effect of that size.

Backup: two-dimensional pairs

Pair	$s_{\min}/s_1$			combined data stat unc. [%]	
	FHC	RHC	COMB		
$\theta_p \times \delta p_T$	0.433	0.332	0.611	13.3%	29–103
$\cos \theta_\pi \times \cos \theta_\mu$	0.245	0.315	0.564	14.9%	33–64
$\theta_{\pi p} \times W_{\pi p}$	0.379	0.214	0.537	78.3%	39–276
$\cos \theta_\pi \times \theta_{\mu\pi}$	0.037	0.133	0.171	25.5%	29–93
$\cos \theta_\mu \times p_\mu$	0.002	0.261	0.106	9.0%	33–57
$\theta_p \times W_{\pi p}$	0.012	0.152	0.112	200%	36–568
$\cos \theta_\pi \times p_\pi$	0.024	0.158	0.052	13.8%	32–62
$\theta_{\pi p} \times \delta p_T$	0.023	0.035	0.027	10.4%	26–52

$\theta_p \times \delta p_T$, FHC+RHC combined

BLINDED: CV PSEUDO-DATA



Dashed frame: open-ended δp_T bin, height = integral / plotted width (δp_T drawn over 0.3–2.5 GeV/c)

- ▶ Two usable pairs, combined only; FHC and RHC responses cannot be inverted.
- ▶ Combining modes doubles the conditioning through sample size and shared flux, not through $\nu/\bar{\nu}$ separation.
- ▶ $\theta_{\pi p} \times W_{\pi p}$ conditions like the usable pairs but carries no information: conditioning and data content must be read together.
- ▶ $\theta_p \times \delta p_T$: closure 1.91/4, cells 0.70–1.36; MCS term $\leq 16\%$, 0.20σ .
- ▶ 2D normalisation agrees with the one-bin total to a median 1.5%.

Backup: consistency between configurations

Combined vs. its inputs

- ▶ Combined truth / fluence-weighted average of FHC and RHC (0.425/0.575): 1.000 (inclusive), 1.001 (proton-tagged).
- ▶ Observable by observable (A_C -smeared): 1.006 (0.981–1.030) and 1.046; the spread is the configuration dependence of A_C .
- ▶ Exposures sum exactly; per-configuration fluxes reproduce the POT-weighted combination to 0.01%.

RHC/FHC

- ▶ Tune 0.901 (incl.), 0.878 (1p); extracted 0.901, 0.873. The deficit is in the input ($\bar{\nu}_\mu$ -dominant sample); the tune ratio is what to compare at unblinding.

NuMI ν_e CC1 π (sibling)

- ▶ Same framework, flux (new G4, fixed geometry), signal topology, systematic suite; FHC Run 2 and Run 4 POT agree exactly.
- ▶ Run 1 FHC: ν_e note quotes 3.283×10^{20} (includes open trigger); here 2.192×10^{20} , the non-open-trigger data.
- ▶ Run 4 overlay = Run4c + Run4d; only one may be used.
- ▶ Pion threshold: $KE > 40 \text{ MeV}$ vs. $p_\pi > 0.175 \text{ GeV}/c$.
- ▶ Opening angle: $\theta_{e\pi} < 170^\circ$ vs. $\theta_{\mu\pi} < 149^\circ$.
- ▶ Regularisation: first vs. second derivative.
- ▶ Beamline unisims: in their covariance, evaluated but outside ours.

Backup: control regions, transfer factors

Transfer factor $T = N^{\text{SR}}/N^{\text{CR}}$ **of each target class**, uncertainty (%)

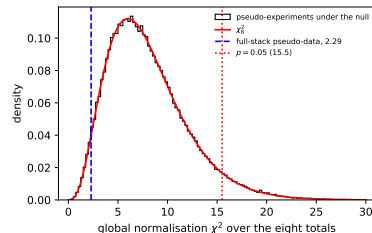
Region		T	flux	xsec	re-int	det.	total
FHC	CC0 π	0.0278	3.3	11.7	6.7	27.1	30.5
	multi- π	2.509	3.0	4.7	2.6	18.5	20.2
	π^0 (CC+NC)	0.0194	2.2	5.8	1.9	14.9	16.8
RHC	CC0 π	0.0286	3.0	11.5	6.6	30.6	33.6
	multi- π	2.453	1.9	4.0	2.8	21.3	22.3
	π^0 (CC+NC)	0.0197	1.7	6.1	1.6	11.8	19.4

Flux cancels (CR–SR correlation 0.98–1.00); detector term largest.
Split-half test: only the CC0 π detector term (27–31% vs. 6–9% floor)
is a genuine sensitivity; the π^0 term sits at its MC-stat floor.

- ▶ **CC0 π** : target purity 72%, 0.6% stat.; validates the CC0 π population, not the proton-as-pion probability. Conditionally eligible for a fitted normalisation.
- ▶ **π^0** : purity 50.1/51.0%; diagnostic first, fit considered only after the pre-fit result is recorded.
- ▶ **multi- π** : $T \approx 2.5$, diagnostic.
- ▶ **cosmic**: validates beam-off to $\approx 60\%$ (21% stat.); EXT is 1.6% of the signal region.
- ▶ Not covered: NC/ ν_e without π^0 , out-of-FV, CC1 π outside phase space ($\approx 30\%$).
- ▶ Direct test of mis-ID (inverted pion-PID region) not defined.

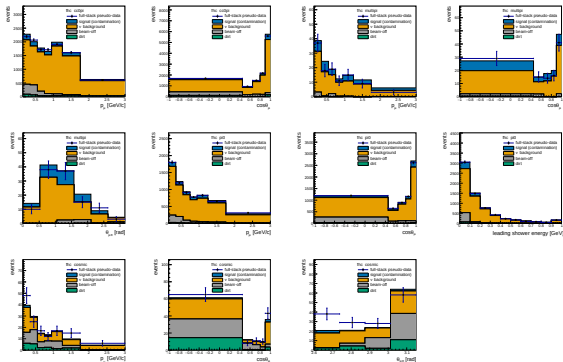
Backup: frozen procedure for the beam-on control regions

1. **Examined:** totals and reco p_μ , $\cos\theta_\mu$, $\theta_\mu\pi$ (leading shower energy for π^0); FHC and RHC separately, combined as cross-check; totals per run period; full prediction ν +EXT+dirt.
2. **Statistics:** per region and mode, normalisation pull; normalisation-marginalised shape χ^2 ; global χ^2 over the 8 totals with the full pre-fit covariance (the pass/fail statistic).
3. **Pass:** global $p > 0.05$ (null: 5.2% fail at $p < 0.05$). $|\text{pull}| > 2$ or shape $p < 0.05$ flags a region.
4. **Responses:** (i) localised failure: reconstruction investigation; (ii) fitted normalisation of the target class, $\text{CC0}\pi$ and π^0 only, if shape passes and LR $p < 0.01$; (iii) otherwise add the discrepancy as an uncertainty via T ; (iv) revise the model before opening the signal region.
5. **Prohibited:** changing cuts or binnings, choosing a generator after inspection, bin-by-bin reweighting, any use of the signal region.



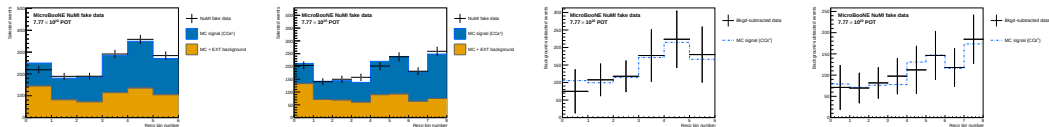
- ▶ Full-stack pseudo-data: global $\chi^2 = 2.29/8$, 16 pulls within ± 0.8 , 22 shape tests $p > 0.12$.
- ▶ Power: one region off by $> 12\%$ ($\text{CC0}\pi$, π^0) or $> 44\text{--}62\%$ (multi- π , cosmic) fails; a common shift only beyond 50% (absorbed by flux and xsec terms).
- ▶ Signal held at the tune: a 30% excess moves $\text{CC0}\pi$ by 2.8%, inside every pull resolution.

Backup: control-region distributions (pseudo-data)

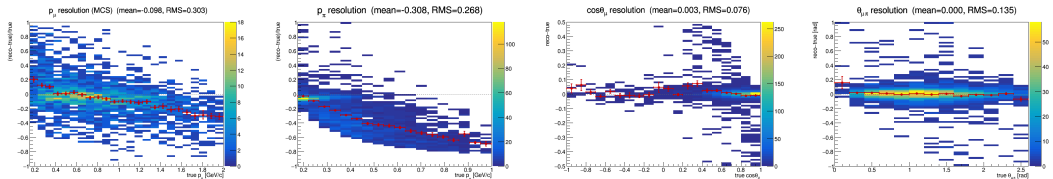


- ▶ FHC, frozen binning; rows: CC0 π , multi- π , π^0 , cosmic.
- ▶ Points are full-stack pseudo-data (ν throw plus independent EXT and dirt throws), in place of beam-on.
- ▶ Data/prediction 0.99–1.05 (FHC), 0.99–1.12 (RHC), 0.99–1.08 (comb.), within statistics.
- ▶ Tests the bookkeeping path of beam-on data, not the background model.
- ▶ Regions overlap only between multi- π and π^0 (38–43% of multi- π).
- ▶ π^0 region without the final multiplicity cut: 11–14% more events, purity and shape unchanged.

Backup: reconstructed spectra and resolution (FHC)

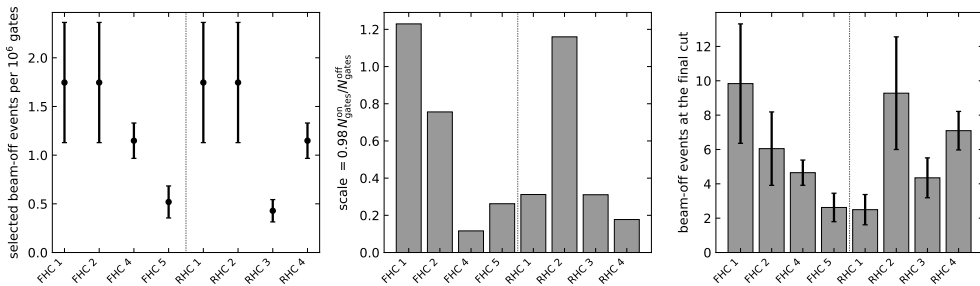


Stage 1 (selected, signal/background split) and stage 2 (background subtracted) for p_μ and $\cos \theta_\mu$.



Resolution: p_μ MCS bias -9.8% , RMS 30%; p_π bias -31% (only -2% in the first bin), RMS 27%; $\cos \theta_\mu$ RMS 0.076; $\theta_{\mu\pi}$ RMS 0.135 rad.

Backup: beam-off normalisation per run period



- ▶ Rate per 10⁶ triggers at the final cut (left), scale 0.98 × on/off triggers (centre), events (right).
- ▶ Run-1 rate is 1.5–4 × the later periods, so samples are matched by period, not pooled.

- ▶ Rates from 8, 14, 40, 10 selected events: Poisson 16–35%.
- ▶ Runs 4–5 beam-off use swtrig-pre (the swtrig flag is zero throughout).
- ▶ Each (beam-off file, run) pair enters each configuration once.

Backup: data release and pipeline

File	Content
curves_<tag>.tsv	per bin: edges, width, unfolded data, stat. error, truth, tune, generators (all A_C -smeared densities)
A_C _<tag>.tsv	row i smeared bin, column j true bin; $\text{smeared}_i = \sum_j A_C[i][j] p_j / w_i$
cov/<tag>/cov_total_plusMCSofficial	covariance
cov/<tag>/cov_<source>.txt	flux, detector, xsec, re-int., POT, targets, MC, EXT, data, MCS
total_xsec.tsv	one-bin totals with breakdown
index_extractions.tsv	every extraction and its status

Tag: <incl|1p>_<FHC5|RHCFULL|COMB>_<obs>; 2D:
incl_COMB_costhpi_costhmu2d, 1p_COMB_thetap_dpt2d.

Reproducibility

- ▶ Result set rebuilt from the processed ntuples with empty caches in one pass.
- ▶ Independent rebuild into a separate tree reproduced every bin of the 51 extractions to 10^{-9} .
- ▶ Universe cache keyed on the normalisation scheme; normalisation manifest audits flux, POT, N_{AR} and unfolding across all configurations.
- ▶ Generator curves reproduced from the released A_C to 4×10^{-8} (96 curves).
- ▶ Source: tag v1.6 (34baa99).

Backup: rejected alternatives (technical supplement, Part III)

- ▶ **First binning optimisation:** seven-bin p_μ failed both criteria (worst diagonal 0.33); five-bin $\cos \theta_\mu$ failed 0.68 narrowly (0.67). Replaced by the binning review.
- ▶ θ_μ **instead of** $\cos \theta_\mu$: same diagonals, less retained information; rank-1 with the uniform stencil.
- ▶ **Pion threshold below 0.175 GeV/c:** resolution 51–56%, efficiency 2.2%, strongly model-dependent correction even as one coarse bin.
- ▶ **Five-bin p_π response:** diagonals 94/37/23/10/6%; pile-up into the first reco bin.
- ▶ **Golden-pion classifier and other p_π estimators:** match the truth ceiling, do not repair the response.
- ▶ **Cosmic-direction cuts:** EXT is a data-driven subtraction (1.6% of the sample); every directional or calorimetric variant removes more signal than the ≈ 5 -event statistical gain.
- ▶ **Six-bin $W_{\pi p}$ and W_{had} :** below both criteria; not reported / reduced to two regions.